Alireza Shafaei

Alireza-Shafaei | shafaee.erfan@yahoo.com | shafaei.erfan.20019@gmail.com | (+98) 991 263 0612 |
Date of Birth: 11 Sep. 2001 | Gender: Male

Education

Sep. 2024 – Current **M.S. in Electrical Engineering (Digital Systems)**, Sharif University of Technology[1]
GPA: 18.02/20 (3.87/4)

Sep. 2019 – Sep. 2024 **B.S. in Electrical Engineering**, Sharif University of Technology
GPA: 17.03/20 (3.52/4)

Feb. 2021 – Sep. 2024 **Minor in Economics**, Sharif University of Technology

Research Interests

- Deep Generative Modeling
- Statistical Machine Learning
- Reinforcement Learning
- Large Language Models & RAG
- Edge-Cloud Computing
- Mathematical Optimization Theory
- Embedded Systems

Selective Courses

- Deep Generative Models (MSc)
- Statistical Learning (MSc)
- Big Data Analytics (MSc)
- Convex Optimization 2 (MSc)
- Deep Learning (MSc)
- Neural Networks (MSc)
- Data communication network (MSc)
- Embedded Systems (MSc)
- Neuroscience of Learning, Memory, Cognition (MSc)
- Adv. Computer Architecture (MSc)
- Intro. to Machine Learning
- Convex Optimization 1
- Communication Circuits
- Electronics 3
- Pulse Tech. and Digital Circuits
- Design Microprocessor Systems
- Econometrics
- Financial Economics
- Digital Economics
- Macroeconomics
- Microeconomics
- System-2 AI (MSc, Audit)
- Deep Reinforcement Learning (MSc, Audit)

Academic Projects & Research Experience

- **LLM-Enabled In-Context Learning for Task Offloading in Edge Networks (2025-present) (master thesis):** Resource allocation as a service (DNN Partitioning) utilizing RAG and Prompt engineering techniques, integrated with mathematical optimization (queueing stability and Lyapunov drift). *Supervisor: Dr. Shah Mansouri.*
- **Reinforcement Learning and Applications (2024)(bachelor thesis):** Using Q-Learning to train agents in continuous environments; reducing resolution by extracting important states. *Supervisor: Dr. Bagheri Shouraki.*
- **Cost-Aware Retrain/Alert/Rollback Framework for Production ML Under Concept Drift (2026–present):** Building the core evaluation harness for a three-stage drift-response policy, covering the prequential loop, cost ledger, retrain gates, canary deployment, and typed rollback under realistic label delay. Work spans data-quality monitoring, cost-based decision policies. *Supervisor: Dr. Khalaj and Dr. Hadibeik.*

Publications

- **Retrain, Alert, or Rollback? A Three-Stage Decision Framework for Production ML Under Concept Drift**
  A. Shafaei, A. Sadeghian, H. Hadibeik, BH. Khalaj
  (Results completed; manuscript in preparation.)

Skills

Programming	Python: (PyTorch, PySpark, Tensorflow, scikit-learn, Django, Keras, Pandas, NumPy, Matplotlib, Seaborn, SciPy, Gym, MLflow), C, Verilog, MATLAB
Design & Sim	PSpice, HSPICE, Proteus, ADS, Omnet++
Other	MIPS/AVR/ARM, Linux, Embedded Systems, LaTeX, Microsoft Office, Problem Solving

Course Projects

- **Generative Models & Deep Learning (2025–2026):**
  - Implemented and trained an Energy-Based Model (EBM) on the MNIST dataset using Langevin dynamics.
  - Developed a time-aware post-training quantization framework for diffusion language models to achieve significant model compression and inference acceleration.
  - Implemented WaveNet (autoregressive generative model for raw audio), Probabilistic Graph Forecasting, CVAE for digit generation, and normalizing flows (RealNVP, PermutFlow).
  - Built a continuous retraining loop using **Prefect** for orchestration and **MLflow** for lifecycle tracking and model registry. Developed a **FastAPI** deployment server featuring dynamic, zero-downtime model hot-swapping. Implemented a **SciPy (KS-Test)** monitoring engine to detect data/concept drift and trigger automated retraining.
- **Advanced Machine Learning & Data Analysis (2024–2025):**
  - *Massive Data Analysis:* Designed graph/collaborative recommenders (Pixie-style, Topic-Specific PageRank, HITS) and built a PySpark Structured Streaming pipeline for real-time sentiment aggregation.
  - *Statistical Learning:* Built end-to-end pipelines for age/gender estimation (Linear/Ridge/Lasso, PCA, ROC-AUC) with a custom drag-and-drop Python GUI for bounding box face detection.
  - Implemented GAN-BERT for AI vs. Human text recognition, complex CNNs for TIG Welding classification, and Hopfield/SOM networks.
- **Networking & Embedded Systems (2024–2025):**
  - *Data Networks:* Implemented a UDP-based DNS server and verified queries in Wireshark; built socket-level SMTP clients and a mailbox-capable SMTP server with auth.
  - *IoT Real-time Event Detection (Orange Pi):* Developed an edge device with LAMP dashboard, Python audio detectors, and MQTT pub/sub to a Mosquitto broker, hardened with systemd limits.
  - Implemented Data Link, MAC, and Network layers in Omnet++.
- **Hardware & Circuit Design (2019–2023):** Designed and simulated MIPS multi-cycle architecture (Verilog), high-frequency MOSFET Amplifiers and Opamp 741 (HSPICE), digital stopwatches, and multivibrators (Proteus/PSpice).
- **Early ML & Computational Modeling (2023–2024):** Hodgkin–Huxley & LIF neuron simulations, VAE/CVAE implementations, clustering with EM Algorithm for GMMs, and classical classification (SVM, Decision Trees, AlexNet Fine-tuning).

Teaching Assistantship

- Neural Networks (MSc) — Dr. Bagheri Shouraki (2025)
- Machine Learning and Vision Lab — Dr. Mohammadzadeh (2025)
- Electronics 1 & 2 — Dr. Fakharzadeh (2022, 2023)
- Principles of Solid State Devices — Dr. Sarvari (2023)

– Design Microprocessor Systems — Dr. Hajsadeghi (2023)

Work Experience

– **AI Engineer & Backend Developer**, Athlx (Startup, 2026)
  – Built scalable **Django** backends for Building Management Systems (BMS) to process real-time IoT telemetry and Developing time-series anomaly detection models for predictive maintenance and facility energy optimization.

Honors & Awards

– Ranked **21st** among 20,000 in Iranian University Entrance Exam for M.Sc. in Electrical Engineering, 2024
– Ranked **24st** among 150,000 in Iranian University Entrance Exam (Math & Physics), 2019

Languages

Persian	Native
English	Advanced
Arabic	Intermediate

References

Dr. Babak Hossein Khalaj | *Professor* | *Sharif University of Technology*
khalaj@sharif.edu

Dr. Saeed Bagheri Shouraki | *Professor* | *Sharif University of Technology*
bagheri-s@sharif.edu

Dr. Hamed Shah-Mansouri | *Assistant Professor* | *Sharif University of Technology*
hamedsh@sharif.edu

Dr. Hamid Hadibeik
hadibeik@gmail.com

[1]Ranked 151-200 in electrical engineering according to QS World University Ranking in 2025. (Ranked 1st in Iran)